Yiwen Chen

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EDUCATION

- **M.Sc., China University of Geosciences (Wuhan)** 2022.09-2025.06

Computer Science **GPA: 92.03125/100 Rank(3/191)**

- **B.S., Beijing Language and Culture University** 2018.09-2022.06

Digital Media Technology **GPA:3.62/4.0 Top 10%**

PUBLICATIONS

Article: Graph Convolutional Network for Water Network Partitioning

As First Author, Accepted, *Expert systems with applications* (**JCR Q1, IF=7.5**)

DOI: https://doi.org/10.1016/j.eswa.2024.125502

Patent: A method for optimizing the layout of sensors in a graph neural network water supply network

As First Author, CN202311667909.8

RESEARCH EXPERIENCE

Summer Research, advised by Prof. Yuke Wang, Rice university 2023.07-2023.08

- Read articles that focus on how to improve the speed and performance of large language model inference such as Gemini, DataStates-LLM, ServerlessLLM, PagedAttention
- Learn GPU programming and CUDA, reproduce mamba-minimal

Graph Convolutional Network for Water Network Partitioning 2023.09-2024.06

- Research project completed independently from 0 to 1.
- By reading a large amount of literature, I found it is essential to utilize the GCN algorithm to combine pressure data and structural topology information comprehensively to cluster the network nodes which nearly nobody has done before.
- Learning materials about deep learning and hydrology, I simulated and generated datasets with Python code.
- On the PyTorch platform, I tried and conducted various experiments about different GCN methods such as Structural Deep Clustering Networks and KMeans on different water networks.
- By analyzing the result, I found a Cross-Attention Fusion Based Enhanced Graph Convolutional Network for Clustering performance well, thus I chose it as my method.
- Write the manuscript by Latex and draw pictures with Visio and Python.

A method for optimizing the layout of sensors in a graph neural network water supply network 2023.09-2024.01

- Use graph convolutional network clustering algorithm to partition the water supply network into sub-regions, and arrange sensors in each region after partitioning, in order to better detect the entire water supply network system.

The 6th/7th Academic Conference on Intelligent Optimization and Scheduling ***Wall Exhibition*** 2024/2023

- Write a review, titled "A review of leak detection and localization methods for complex water supply systems"
- Classify and deeply analyze the literature from the perspectives of physical sensors, data-driven and model-driven for the current important research results in this field, and discuss the directions that can be studied in the future
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The 15th Guangdong Hong Kong Macao Greater Bay Area Doctoral Student Nanshan Academic Forum and Tsinghua University 706th Doctoral Student Academic Forum ***Wall Exhibition*** 2023.06

AI Speech Recognition Research 2020.04-2021-04

Leader of College Students' Innovative Entrepreneurial Training Program, Beijing Municipal Level

- Building and training a Chinese tone recognition model and published a simple speech recognition mini program.
- On the PyTorch platform, we extracted tone information features from the Tsinghua 30h dataset, and established a CNN+CTC model through VScode's remote SSH to connect remote service to use GPU for tone recognition training.
- Developed a speech recognition mini Wechat program by learning open-source code.
- Gained knowledge in areas such as Speech Signal Processing, Fourier Transform, Neural Networks, and Mini program development

English Learning System Based on Virtual Reality 2021.11-2022.04

Undergraduate graduation project

- Use C#, SteamVr, Unity, VRTK and Vive Pro to build an English learning system
- I realize the functions of word collection and statistics, character interaction and dialogue, etc
- Users can use handles to interact with the objects and NPCs in the virtual world

  Can be visited by https://youtu.be/-ZioAZY1JSs

ADDITIONAL EXPERIENCE

University of California Berkeley Summer Visit 2023.07-2023.08

- Learning Data Structures and Algorithms CS61b Course in EECS and Marketing class in Haas
- Know how to use java to produce game 2048 and use java to realize a simple version of Git

Summer School of Peking University 2022.07-2022.08

- Learning Elective Course on Intelligent Computing Systems, scored 93

Volunteer for the 3rd Youth Academic Seminar of the Cross Science Department of the *National Natural Science Foundation of China* 2023.10

Head of the Academic Department of the Graduate Association of the School of Computer Science 2022

HONORS & AWARDS

- A member of Baidu AI Pinecone Talent Development Elite Class 2024
- Successfully passed the WPS office C++ back-end school-enterprise cooperation training camp 2024
- Academic Scholarship (First Class) for three consecutive years 2022-2024
- High School Information Technology Teaching Qualification Certificate 2023
- Speaker on behalf of the incoming graduate student 2022
- Outstanding youth league cadres at the school level 2022/2018
- S Award in the American Mathematical Modeling Competition for College Students 2021
- Merit student of Beijing language and culture university 2021

PROFESSIONAL SKILLS

Good at Mathematics:

- Second Prize in the Beijing Division of the 12th National College Student Mathematics Competition (2020)
- Second prize in the Hubei Division of the National Mathematics/ Chemistry / Biology Olympiad(2017)

Programming: C/C++/Python/JAVA/C #/Matlab/CUDA, master Object-Oriented programming concepts.

Tools: Git/CMake/Gdb/Vim/Springboot/Qt/Unity/3D-Max

Data Analysis&Visualization: CSS/Javascript/Vue/Html/Echarts/SPSS

Good English Ability: IELTS 7/ Test for English Major-4

PERSONAL SUMMARY

- I enjoy technical communication and often browse and learn on websites such as Blogs, Zhihu, Github, Bilibili.
- Passionate about learning, honest and upright, with the ability to conduct independent research.
- Love life and sports, enjoy dancing, humble and eager to learn, diligent and hardworking, with strong self-learning

ability and ambition, as well as strong teamwork and communication skills.